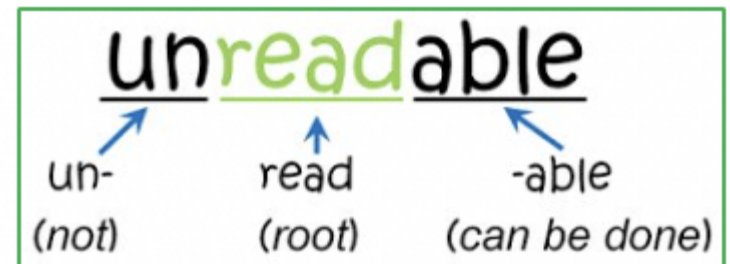
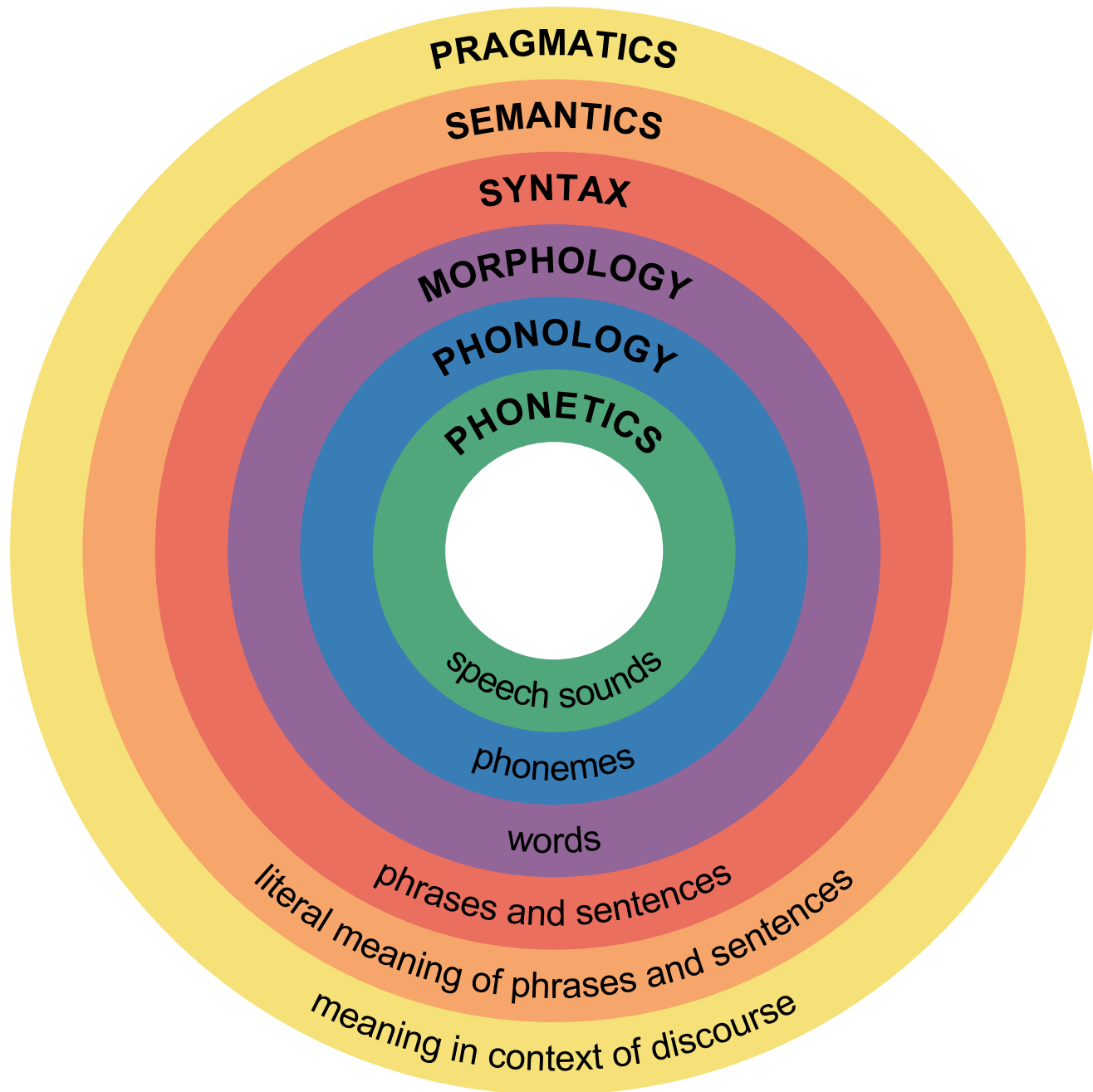


Basic Text Processing: Words & Morphology (Natural Language Processing)

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What is a Text Corpus?

- Corpus is a collection of written texts, especially the entire works of a particular author or a body of writing on a particular subject.
- In linguistics, a corpus or text corpus is a large and structured set of texts.
 - Google Books Ngram Corpus
 - American National Corpus - 22 million words of written and spoken data
 - COBUILD corpus - 4.5 billion words
 - British National Corpus - 100-million-word
 - Corpus of Contemporary American English (COCA) - 425 million words
 - UCSC 10M words Sinhala Corpus – 10 million words

Words

- How many words are in the following sentence?

They picnicked by the pool, then they lay back on the grass and looked at the stars

Words

- How many words are in the following sentence?

They picnicked by the pool, then they lay back on the grass and looked at the stars.

Words/Tokens =

Unique/Distinct words =

Words

- How many words are in the following sentence?

They picnicked by the pool, then they lay back on the grass and looked at the stars.

Words/Tokens =

Unique/Distinct words =

Word Counting in Corpora

- After corpus preparation, additional decisions

- Ignore capitalization at beginning of sentence? Is “**They**” the same word as “**they**”?
- Ignore other capitalization? is “**Company**” the same word as “**company**”
- Stemming? Is “**cat**” the same word as “**cats**”

- Terminology for word occurrences:

- Tokens – the total number of words
- Distinct Tokens (sometimes called word types) – the number of distinct words, not counting repetitions
- The following sentence from the Brown corpus has 17 tokens and 15 distinct tokens:

They picnicked by the pool, then they lay back on the grass and looked at the stars.

Word Frequencies

- Count the number of each token appearing in the corpus (or sometimes single document)
- A frequency distribution is a list of all tokens with their frequency, usually sorted in the order of decreasing frequency
 - Used to make “word clouds”
 - Create your own word cloud?
<https://www.wordclouds.com/>
- Used for comparison and characterization of text



How many words in a corpus?

- Let **N** be the number of tokens
- Let **V** be the size of the vocabulary (the number of distinct tokens)
- Church and Gale (1990) suggest that the **V** grows with at least the square root of the **N** (i.e. $|V| > O(N^{1/2})$).

	Tokens= N	Types= V
Switchboard phone conversations	2.4 million	20 thousand
Shakespeare	884,000	31 thousand
GoogleN-grams	1 trillion	13 million

Corpus Statistics

Corpus Statistics	News French	Commentary English	News German	Commentary English	UCSC-ICTA Sinhala	Tamil
Sentence Pairs	25K		25.5 K		25 K	
Total Number of Words (T)	427 K	351 K	305 K	306 K	252 K	219 K
Vocabulary Size (V)	24 K	21 K	33 K	20 K	37 K	53 K
V/T (%)	5.6	5.9	10.8	6.5	14.6	24.2
Average Number of Words per Sentence	17	14	12	12	10	9
Corpus Statistics	Europarl Finnish	Corpus English	Europarl Hungarian	Corpus English	Europarl Czech	Corpus English
Sentence Pairs	25K		25 K		25 K	
Total Number of Words (T)	205 K	247 K	238 K	267 K	249 K	282 K
Vocabulary Size (V)	34 K	14 K	34 K	13 K	29 K	12 K
V/T (%)	16.8	5.6	14.5	5.1	11.7	4.6
Average Number of Words per Sentence	8	10	10	11	10	11

Corpus Statistics

Corpus Statistics	EN	FR	DE	SI	TA	FI	HU	CZ
Total no. of Words (T)	36.5M	42M	33.8M	10.1M	4.2M	33.7M	12.4M	12.9M
Vocabulary (V)	102K	121K	331K	449K	400K	677K	312K	173K
V/T (%)	0.28	0.29	0.98	4.4	9.5	2.0	2.5	1.3

Preliminary Text Processing Required:

- Find the words:

- Filter out 'junk data'

- Formatting / extraneous material
 - First be sure it doesn't reveal important information

- Deal with upper / lower case issues

- Tokenize

- Decide how you define a 'word'
 - How to recognize and deal with punctuation
 - Apostrophes (one word it's vs. two words it's)
 - Hyphens (snow-laden vs. New York-New Jersey)
 - Periods (kept with abbreviations vs. separated as sentence markers)

Preliminary Text Processing Required:

- Word segmentation
 - No white space in Japanese language
 - Compound words –
 - “Lebensversicherungsgesellschaftsangestellter”
- Additional issues if OCR'd data or speech transcripts
- Morphology (To stem or not to stem?)
 - Depends on the application

Exercise 1 – Text Preprocessing

Normalize the following text:

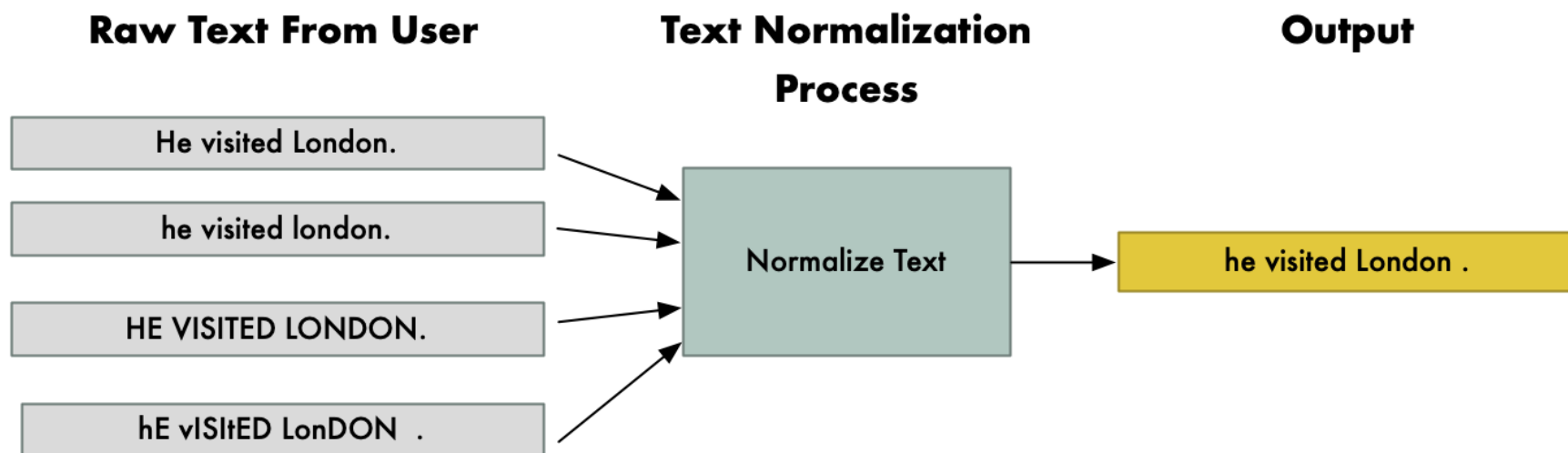
- u r amazing!!!
- Thx for ur help :)
- pls send d doc ASAP
- nooooooo wayyyy!!!
- kohomd oyata
- llovetoreadbooks
- Thisisaveryhardproblem

Basic Text Processing

- Every NLP task needs to do text normalization to determine what are the words of the document:
 - Segmenting/tokenizing words in running text
 - Special characters like hyphen “-” and apostrophe ‘
 - Normalizing word formats
 - (Non) capitalization of words
 - Reducing words to stems or lemmas
 - Segmenting sentences in running text
- To do these tasks, we need to use morphology

Text Normalization

- Text normalization reduces variations in word forms to a common form when the variations mean the same thing.



Text Normalization

- Text normalization reduces variations in word forms to a common form when the variations mean the same thing.
- Stemming and Lemmatization:
 - **Stemming** just removes or stems the last few characters of a word, often leading to incorrect meanings and spelling.
 - **Lemmatization** considers the context and converts the word to its meaningful base form, which is called Lemma.

Stemming VS Lemmatization

Word	Stemming	Lemmatization
Information		
Informative		
Computers		
Feet		
Went		

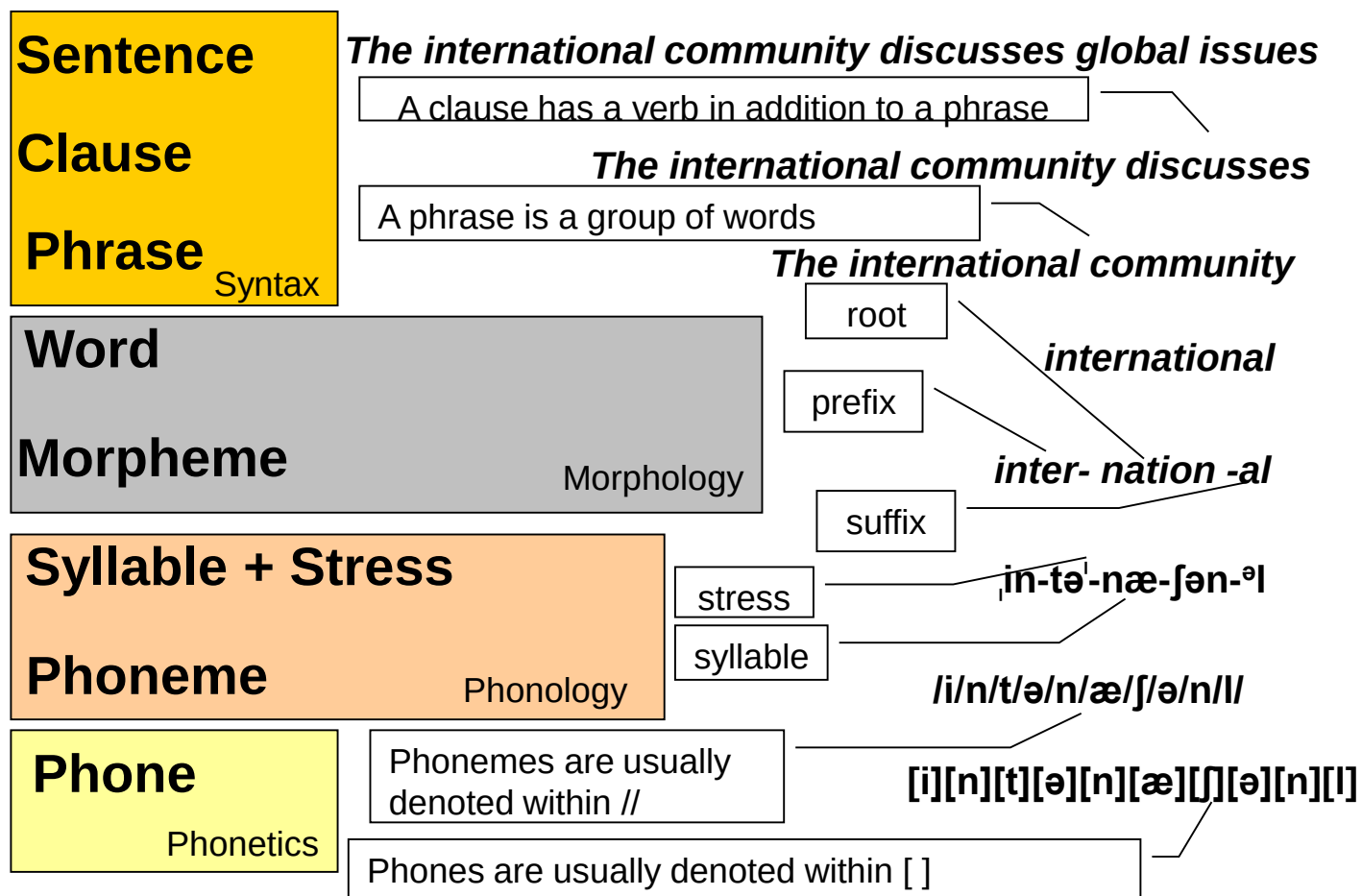
Stemming VS Lemmatization

Word	Stemming	Lemmatization
Information	Inform	Information
Informative	Inform	Information
Computers	Comput	Computer
Feet	Feet	Foot
Went	We	Go

Word!

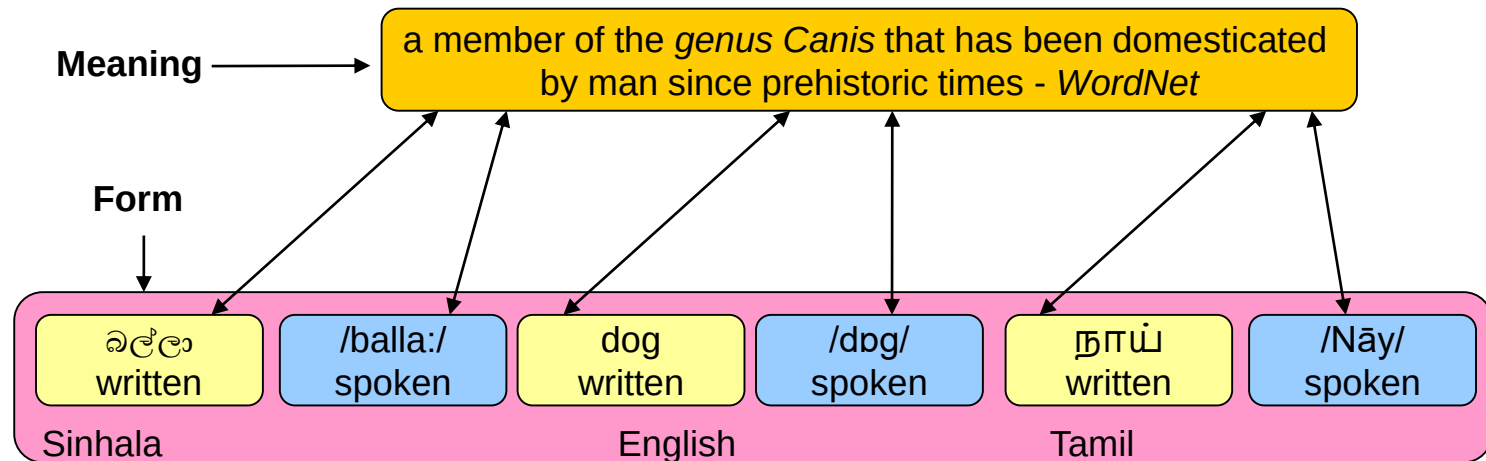
- In formal languages, words are arbitrary strings
- In natural languages, words are made of meaningful sub units called morphemes
- How we define a word?

Word in the Context of Linguistic Analysis



What is a Word?

- A word is a unique linguistic entity, which can be expressed in either speech or writing - *word has a form*
 - **Spoken vs. Written**
- A word is a linguistic device that relates *form* and *meaning*
- The relation between *form* and *meaning* is arbitrary
 - The concept DOG has arbitrary spoken and written forms in different languages



Lexical Semantics

- The *meaning* of a word is largely determined by its context
- Consider these two uses of the word *bank*
 - Instead, a *bank* can hold the investments in a custodial account in the client's name
 - But as agriculture burgeons on the east *bank*, the river will shrink even more
- These two contextual variations of the meaning *bank* are called (*word*) *senses*
- Therefore, a sense is an aspect of the meaning of a word, usually denoted by *bank*¹ and *bank*²

Relations between Senses

- When the meaning of two senses of two different words are identical or nearly identical they are called **synonyms**.
 - couch/sofa, vomit/throw up, car/automobile
- Word with opposite meanings are **antonyms**
 - long/short, big/little, fast/slow, cold/hot, dark/light, rise/fall, up/falls
- One sense is a **hyponym** of another sense if the first sense is more specific, denoting a subclass of another
 - car is a hyponym of vehicle, dog is a hyponym of animal
- One sense is a **hypernym** of another sense if the first sense is more general, denoting a subclass of another
 - vehicle is a hypernym of car , animal is a hypernym of dog

Relations between Senses

- One sense is a **meronym** of another sense if the first sense is part of the second
 - leg is a *meronym* of chair, wheel is a meronym of of car
- One sense is a **holonym** of another sense if the first sense has *a part of* denoted by the second,
 - Chair is a *holonym* of leg, car is a *holonym* of wheel

Comparison of *Meaning*, *Written* and *Spoken* Forms

Meaning	Written Form	Spoken Form	Name	Examples
Different	Different	Different	Different	cat,dog
Different	Different	Same	Homophones	bear , bare
Different	Same	Different	Homographs	bass- fish, bass- music
Different	Same	Same	Homonyms	bank
Same	Different	Different	Synonyms	high, tall
Same	Different	Same	Orthographic Variants	labor, labour
Same	Same	Different	Phonetic Variants	either /iy dh er/ , /ay dh er/
Same	Same	Same	Identical	-

Exercise: Identify Sinhala/Tamil examples related to the above table

Morphology

- **Morphology** is the field of linguistics that studies the internal structure of words
- Study of the rules that govern the combination of morphemes.
- How words are built up from smaller meaningful units called morphemes
- morph = shape, Logy = logos = study of

Morphology

- General morphological theory applies to all languages as all natural human languages have systematic ways of structuring words (even sign language)
- Must be distinguished from the morphology of a specific language
 - English words are structured differently from German words, although both languages are historically related
 - Both are vastly different from Arabic

Morphemes

- Smallest units of meaning
- Express concepts or relationships
 - Ex: car, table, anti-, re-.
- Express syntactic features
 - number (singular, plural)
 - tense (present, past, future)
 - gender (masculine, feminine)
 - case (nominative, accusative, genitive, dative, locative, ablative, instrumental, vocative) .

Morphemes

- Morph:

- Morphemes as parts of a word.

- Car – the morpheme *car* is realized as the morph *car* to form the word *car*.
 - Cars – the morpheme *car* and the plural morpheme are realized as *car* and +s respectively, to form the word *cars*.

- Allomorphs:

- The different forms of a morpheme.

- Ex: the plural morpheme in English has several allomorphs (+es, +s, stem vowel alteration, etc.).
 - Ex: take, took.

Morphemes: Types

- Free morphemes
 - Can form words by themselves.
 - Ex: Car, dog.
- Bound morphemes
 - Must be combined with other morphemes to form words.
 - Ex: Plural morpheme, anti-.
- Words can be formed by free morphemes only, or free and bound morphemes.

Question — Identify Free and Bound Morphemes

- unhappy
- cats
- teacher
- impossible
- Nationalization
- Replay
- Kindness
- disagreement

Morphemes: 2 classes

- We can usefully divide morphemes into two classes
 - Stems: The core meaning bearing units
 - Affixes: Bits and pieces that adhere to stems to change their meanings and grammatical functions

Affixes

- Prefixes appear in front of the stem to which they attach
 - **un-** + *happy* = *unhappy*
- Infixes appear inside the stem to which they attach
 - **blooming-** + *absolutely* = *absobloominglutely*
- Suffixes appear at the end of the stem to which they attach
 - *Happy* + **-ness** = *Happiness*
 - English may stack up to 4 or 5 suffixes to a word
 - Agglutinative languages like Turkish may have up to 10
- Circumfixes appear at both the beginning and end of stem
 - German past participle of *sagen* is *gesagt*: **ge-** + *sag* + **-t**
- Spelling and sound changes often occur at the boundary
 - Very important for NLP

Two Broad Classes of Morphology

- Inflectional Morphology
- Derivational Morphology

Inflection

- Inflection modifies a word's form in order to mark the grammatical subclass to which it belongs
 - *apple* (singular) > *apples* (plural)
- Inflection does not change the grammatical category (part of speech)
 - *apple* – noun; *apples* – still a noun
- Inflection does not change the overall meaning
 - both *apple* and *apples* refer to the fruit

Think examples in your own language!

Derivation

- Derivation creates a new word by changing the category and/or meaning of the base to which it applies
 - *create* + **-tion** = **creation**
- Derivation can change the grammatical category (part of speech)
 - *sing* (verb) $\neq$ **singer** (noun)
- Derivation can change the meaning
 - *judge* (form an opinion) $\neq$ **judgment** (ability to make considered decisions)
- Derivation is often limited to a certain group of words
 - You can **Clintonize** the government, but you can't **Bushize** the Government
 - 'mahindakaranaya'?
 - This restriction is partially phonological

Inflection & Derivation: Order

- **Order is important** when it comes to inflections and derivations
 - **Derivational suffixes must precede inflectional suffixes**
 - *sing* + *-er* + *-s* is ok
 - *sing* + *-s* + *-er* is not
 - This order may be used as a clue when working with natural language text

Inflection & Derivation in English

- English has few inflections
 - Many other languages use inflections to indicate the role of a word in the sentence
 - Use of case endings allows fairly free word order
 - English instead has a fixed word order
 - Position in the sentence indicates the role of a word, so case endings are not necessary
 - This was not always true; Old English had many inflections
- English has many derivational affixes, and they are regularly used to form new words
 - Part of this is cultural -- English speakers readily accept newly introduced terms

Inflections in English

- examples from J&M, sections 3.1 – 3.3 (2nd ed.)

Morphological Form Classes	Regularly Inflected Verbs			
stem	walk	merge	try	map
-s form	walks	merges	tries	maps
-ing participle	walking	merging	trying	mapping
Past form or -ed participle	walked	merged	tried	mapped

Questions

- Write down 5 examples of inflection in the Sinhala/Tamil language.

Derivations in English

A very common kind of derivation in English is the formation of new nouns, often from verbs or adjectives. This process is called **nominalization**. For example, the suffix *-ation* produces nouns from verbs ending often in the suffix *-ize* (*computerize* → *computerization*). Here are examples of some particularly productive English nominalizing suffixes.

Suffix	Base Verb/Adjective	Derived Noun
-ation	computerize (V)	computerization
-ee	appoint (V)	appointee
-er	kill (V)	killer
-ness	fuzzy (A)	fuzziness

Adjectives can also be derived from nouns and verbs. Here are examples of a few suffixes deriving adjectives from nouns or verbs.

Suffix	Base Noun/Verb	Derived Adjective
-al	computation (N)	computational
-able	embrace (V)	embraceable
-less	clue (N)	clueless

- examples from J&M, sections 3.1 – 3.3 (2nd ed.)

Questions

- Write down 5 examples of derivation in the Sinhala/Tamil language.

Question

- Identify the stems, derivational morphemes and inflectional morphemes separately of all the following words.

Word
establishment
repaired
unfortunately
wolves
mismanagement

Classes of Words

- **Closed** classes are fixed – new words cannot be added
 - Pronouns, Prepositions, Comparatives, Conjunctions, Determiners (Articles and Demonstratives).
 - Function words.
- **Open** classes are not fixed – new words can be added
 - Nouns, Verbs, Adjectives, Adverbs.
 - Content words.
 - New content words are a constant issue for NLP.

Write down 5 examples for each

Creation of New Words

- **Derivation** - adding prefixes or suffixes to form a new word
 - *Clinton* -> *Clintonize*
- **Compounding** - combining two existing words
 - *home* + *page* -> *homepage*
- **Clipping** - shortening a polysyllabic word
 - *Internet* -> *net*
 - *Examination* → *exam*
- **Acronyms** - take initial sounds or letters to form new word
 - *Unesco* -> *United Nations Educational, Scientific and Cultural Organization*
- **Blending** - combine parts of two words
 - *motor* + *hotel* -> *motel*
 - *smoke* + *fog* -> *smog*
- **Backformation**
 - *resurrection* -> *resurrect*
 - *Editor* -> *Edit*

Word Formation Rules: Agreement

- Plurals

- In English, the morpheme “s” is often used to indicate plurals in nouns.
- Nouns and Verbs must agree in plurality.

- Gender – nouns, adjectives and sometimes verbs in many languages are marked for gender

- 2 genders (masculine and feminine) in Romance languages like French, Spanish, Italian.
- 3 genders (masc, fem, and neuter) in Germanic and Slavic languages
- More are called noun classes – Bantu has up to 20 genders.
- Gender is sometimes explicitly marked on the word as a morpheme, but sometimes is just a property of the word.

How does NLP make use of morphology?

- Stemming

- Strip prefixes and / or suffixes to find the base root, which may or may not be an actual word
 - Spelling corrections are not made

- Lemmatization

- Strip prefixes and / or suffixes to find the base root, which will always be an actual word
 - Spelling corrections are crucial
 - Often based on a word list, such as that available at WordNet

- Morphological Parsing

- Knowledge of morphemes for a particular language can be a powerful aid in guessing the part of speech and grammatical features for even unknown terms

Stemming

- Removal of affixes (usually suffixes) to arrive at a base form that may or may not necessarily constitute an actual word
- Continuum from very conservative to very liberal modes of stemming
 - **Very Conservative**
 - Remove only plural –s
 - **Very Liberal**
 - Remove all recognized prefixes and suffixes

*for example compressed
and compression are both
accepted as equivalent to
compress.*



for exampl compress and
compress ar both accept
as equival to compress

Porter Stemmer

- Popular stemmer based on work done by Martin Porter
 - *M.F. Porter. An algorithm for suffix stripping. 1980, Program 14(3), pp. 130-137.*
- Very liberal step stemmer with five steps applied in sequence
 - See example rules on next slide
- Probably the most widely used stemmer
 - Has been incorporated into a number of Information Retrieval systems
- Does not require a lexicon.
- Open source software available for almost all programming languages.

Rules of Porter Stemmer

Step 1a

sses	→ ss	caresses	→ caress
ies	→ i	ponies	→ poni
ss	→ ss	caress	→ caress
s	→ ∅	cats	→ cat

Step 1b

(*v*)ing	→ ∅	walking	→ walk
		sing	→ sing
(*v*)ed	→ ∅	plastered	→ plaster
...			

Step 2 (for long stems)

ational	→ ate	relational	→ relate
izer	→ ize	digitizer	→ digitize
ator	→ ate	operator	→ operate
...			

Step 3 (for longer stems)

al	→ ∅	revival	→ reviv
able	→ ∅	adjustable	→ adjust
ate	→ ∅	activate	→ activ
...			

[Stanford - Word Normalization and Stemming](#)

[Jurafsky NLP Coursera Slides](#)

Lemmatization

- Removal of affixes (typically suffixes),
- But the goal is to find a base form that does constitute an actual word
- Example:
 - *parties* -> remove -es, correct spelling of remaining form -> *party*
- Spelling corrections are often rule-based
- May use a lexicon to find actual words

Stemming

adjustable → adjust
formality → formaliti
formaliti → formal
airliner → airlin △

Lemmatization

was → (to) be
better → good
meeting → meeting

Morphological Parsing

- English is continuously gaining new words on a daily basis
- And new words are a problem for many NLP systems
 - New words won't be found in the MRD or lexicon, if one is used
- How might morphology be used to help solve this problem?

Problem of Ambiguous Affixes

- Some affixes are ambiguous:

- -er

- Derivational: Agentive -er Verb + -er > noun (*sing* + -er -> *singer*)
 - Inflectional: Comparative -er Adjective + -er > Adjective (*big* + -er -> *bigger*)

- -s or -es

- Inflectional: Plural Noun + -(e)s > Noun (*cat* + -s -> *cats*)
 - Inflectional: 3rd person sing. Verb + -(e)s > Verb (*write* + -s -> *writes*)

- -ing

- Inflectional: Progressive Verb + -ing > Verb *he is swimming*
 - Derivational: “act of” Verb + -ing > Noun *swimming is good for health*
 - Derivational: “in process of” Verb + -ing > Adjective *swimming pool*

This morphological ambiguity creates a problem for NLP

Complex Morphology

- Some languages requires complex morpheme segmentation
 - In Turkish,
 - Uygarlastiramadiklarimizdanmissinizcasina
 - `(behaving) as if you are among those whom we could not civilize`
 - Uygar `civilized` + las `become`
 - + tir `cause` + ama `not able`
 - + dik `past` + lar `plural`
 - + imiz `p1pl` + dan `abl`
 - + mis `past` + siniz `2pl` + casina `as if`

Computational Morphology

- Analysis (words → encoded meaning)
 - Take a sequence of characters as input, and produce an analysis of the information encoded in the characters.
 - Ex: *Plays* -> (play/noun/plural) or (play/verb/3rd person/singular/present)
- Generation (meaning → words)
 - Generate words from a set of features.
 - Ex: (run/verb/1st person/singular/past) -> *ran*